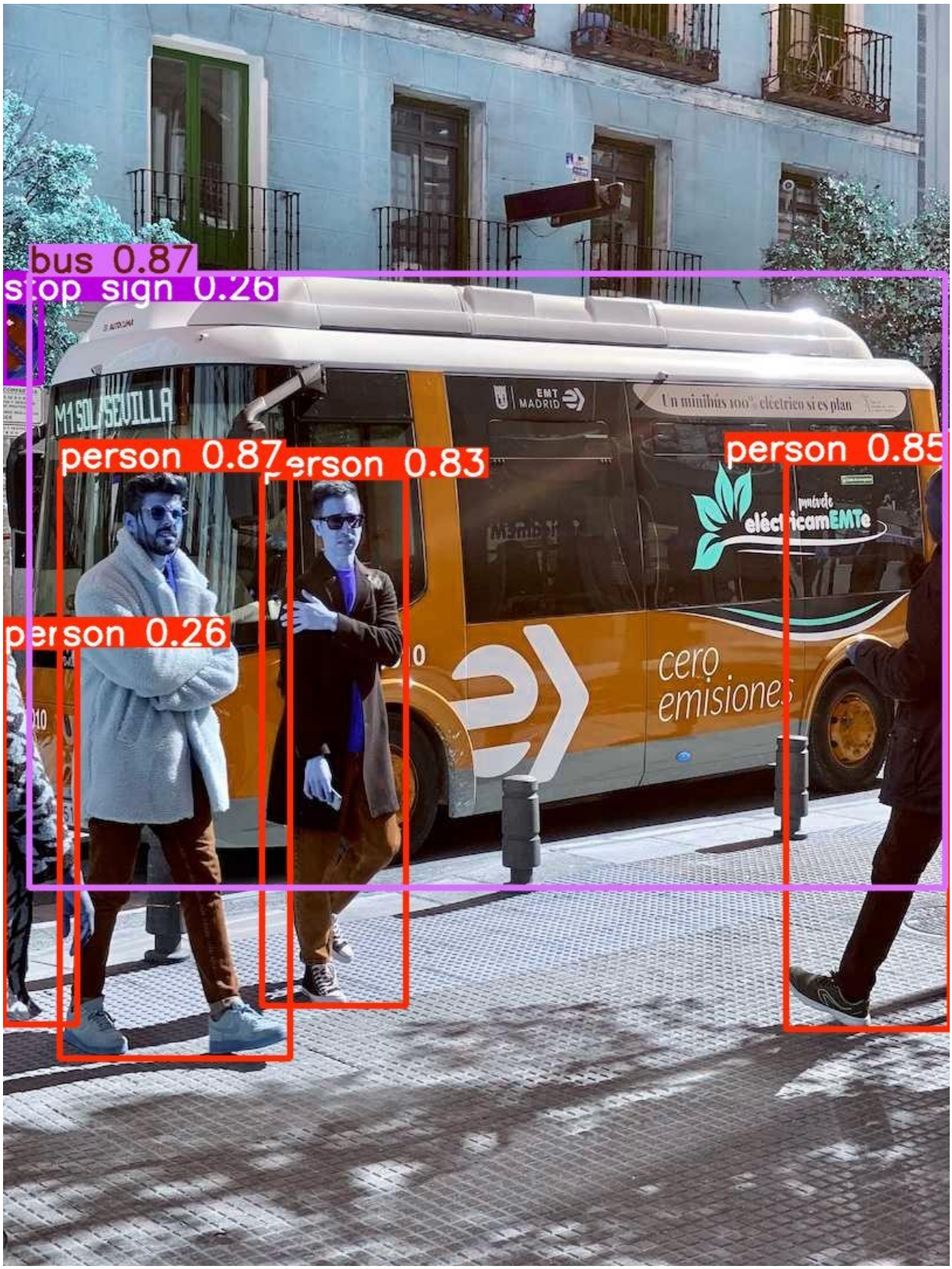
```python
from ultralytics import YOLO
from IPython.display import display
from PIL import Image
model = YOLO("yolov8n.pt")
results = model("https://ultralytics.com/images/bus.jpg")
img = results[0].plot()
display(Image.fromarray(img))
print("\n--- Detected Objects ---")
for result in results:
    for box in result.boxes:
        cls_id = int(box.cls[0])
        conf   = float(box.conf[0])
        label  = model.names[cls_id]
        print(f"Object: {label:<15} Confidence: {conf:.2f}")
import urllib.request
from ultralytics import YOLO
from IPython.display import display
from PIL import Image
url       = "https://ultralytics.com/images/zidane.jpg"
save_path = "C:/Users/sreev/Downloads/test_img.jpg"
urllib.request.urlretrieve(url, save_path)
print(f"✓ Image downloaded to: {save_path}")
model   = YOLO("yolov8n.pt")
results = model(save_path)
img = results[0].plot()
display(Image.fromarray(img))
print("\n--- Detected Objects ---")
for result in results:
    for box in result.boxes:
        cls_id = int(box.cls[0])
        conf   = float(box.conf[0])
        label  = model.names[cls_id]
        print(f"Object: {label:<15} Confidence: {conf:.2f}")
```

```
/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to
'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 7.6MB/s 0.8s.7s<0.1ss.7s

/images/bus.jpg to 'bus.jpg': 100% ━━━━━━━━━━━━ 134.2KB 3.2MB/s 0.0s
image 1/1 C:\Users\sreev\Downloads\bus.jpg: 640x480 4 persons, 1 bus,
1 stop sign, 87.1ms
Speed: 4.2ms preprocess, 87.1ms inference, 1.7ms postprocess per image
at shape (1, 3, 640, 480)
```


bus 0.87
stop sign 0.26
person 0.87
person 0.83
person 0.85
person 0.26
M1 SOL/SEVILLA
EMT MADRID
cero emisiones

```
--- Detected Objects ---
Object: bus             Confidence: 0.87
Object: person          Confidence: 0.87
Object: person          Confidence: 0.85
Object: person          Confidence: 0.83
Object: person          Confidence: 0.26
Object: stop sign       Confidence: 0.26
✓ Image downloaded to: C:/Users/sreev/Downloads/test_img.jpg

image 1/1 C:\Users\sreev\Downloads\test_img.jpg: 384x640 2 persons, 1
tie, 81.8ms
Speed: 3.2ms preprocess, 81.8ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 640)
```

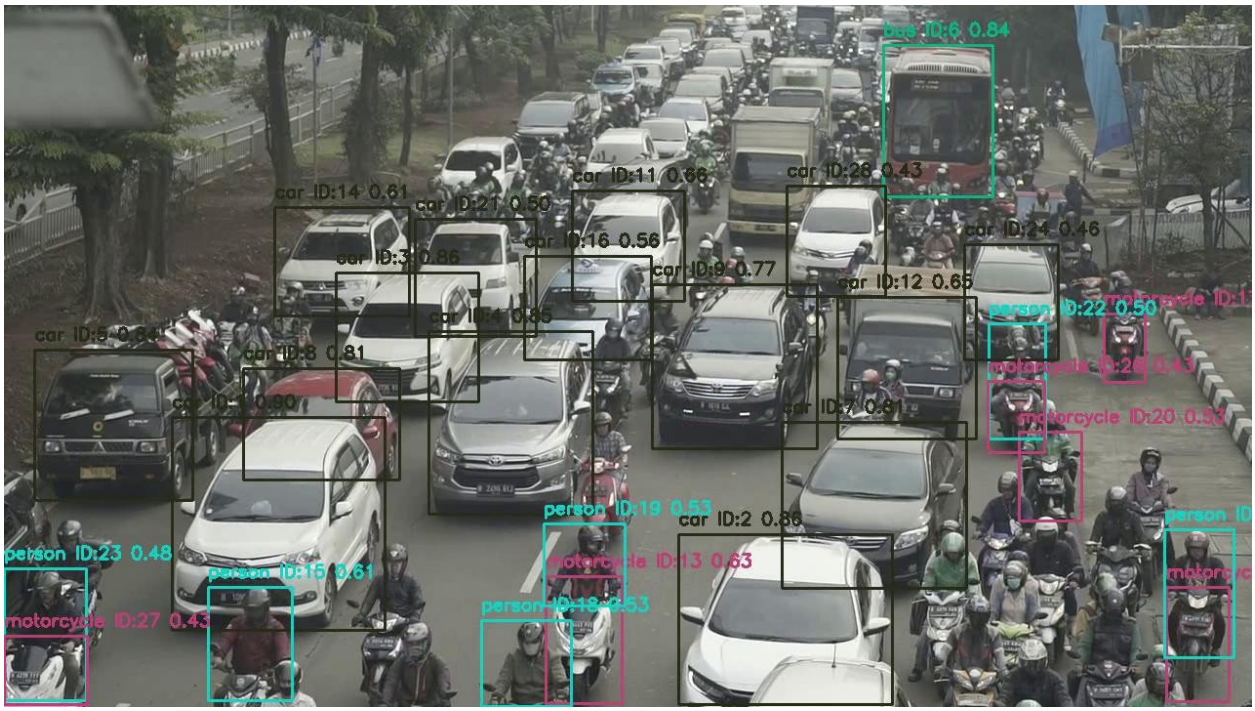
```
--- Detected Objects ---
Object: person          Confidence: 0.84
Object: person          Confidence: 0.82
Object: tie             Confidence: 0.29
```

```python
import cv2
import random
import os
from ultralytics import YOLO
from IPython.display import Video

os.chdir(r"C:\Users\sreev\Downloads")
```

```
model = YOLO("yolov8s.pt")

video_path = r"C:\Users\sreev\Downloads\t1.mp4"
cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print("□ Error: Could not open video. Check the path!")
else:
    print("✓ Video opened successfully!")

# Get video properties
fps = int(cap.get(cv2.CAP_PROP_FPS))
w   = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
h   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
print(f"✓ Video Info → FPS: {fps}, Width: {w}, Height: {h}")

out = cv2.VideoWriter(
    "output.mp4",
    cv2.VideoWriter_fourcc(*'mp4v'),
    fps,
    (w, h)
)
def get_color(cls_id):
    random.seed(cls_id)
    return tuple(random.randint(0, 255) for _ in range(3))
MAX_FRAMES  = 100
frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret:
        break
    if frame_count >= MAX_FRAMES:
        print(f"\n✓ Reached limit of {MAX_FRAMES} frames.
Stopping...")
        break
    results = model.track(frame, stream=True, verbose=False)
    for result in results:
        for box in result.boxes:
            if box.conf[0] > 0.4:   # Only show confidence > 40%
                x1, y1, x2, y2 = map(int, box.xyxy[0])
                cls      = int(box.cls[0])
                conf     = float(box.conf[0])
                track_id = int(box.id[0]) if box.id is not None else 0

                # Draw bounding box
                color = get_color(cls)
                cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)

                # Draw label: class name + track ID + confidence
```

```
                label = f"{result.names[cls]} ID:{track_id}
{conf:.2f}"
                cv2.putText(frame, label, (x1, y1 - 10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 2)
    out.write(frame)
    frame_count += 1

    if frame_count % 10 == 0:
        print(f"Processed {frame_count}/{MAX_FRAMES} frames...")
cap.release()
out.release()
print(f"\n✓ Done! Output saved to: C:\\Users\\sreev\\Downloads\\
output.mp4")
import cv2
from IPython.display import display
from PIL import Image
output_video_path = r"C:\Users\sreev\Downloads\output.mp4"
cap = cv2.VideoCapture(output_video_path)
cap.set(cv2.CAP_PROP_POS_FRAMES, 50)
ret, frame = cap.read()

if ret:
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    img       = Image.fromarray(frame_rgb)
    snapshot_path = r"C:\Users\sreev\Downloads\yolo-video-op.png"
    img.save(snapshot_path)
    print(f"✓ Snapshot saved to: {snapshot_path}")
    display(img)
else:
    print("□ Could not read frame. Check if output.mp4 was created.")
cap.release()
```

```
✓ Video opened successfully!
✓ Video Info → FPS: 50, Width: 1280, Height: 720
Processed 10/100 frames...
Processed 20/100 frames...
Processed 30/100 frames...
Processed 40/100 frames...
Processed 50/100 frames...
Processed 60/100 frames...
Processed 70/100 frames...
Processed 80/100 frames...
Processed 90/100 frames...
Processed 100/100 frames...

✓ Reached limit of 100 frames. Stopping...

✓ Done! Output saved to: C:\Users\sreev\Downloads\output.mp4
✓ Snapshot saved to: C:\Users\sreev\Downloads\yolo-video-op.png
```


bus ID:6 0.84
car ID:14 0.61
car ID:11 0.66
car ID:26 0.43
car ID:21 0.60
car ID:16 0.56
car ID:24 0.46
car ID:3 0.86
car ID:9 0.77
car ID:12 0.65
car ID:8 0.81
person ID:22 0.50
car ID:7 0.61
motorcycle ID:20 0.55
person ID:19 0.53
car ID:2 0.86
person ID:23 0.48
motorcycle ID:13 0.63
person ID:15 0.61
motorcycle ID:27 0.43
person ID:18 0.53